Intermediate Value Theorem

The Proof.

Theorem 8.3. Intermediate Value Theorem

Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$.

Suppose $\alpha \in \mathbb{R}$ is in between $f(a)$ and $f(b)$

(i.e., $f(a) < \alpha < f(b)$ or $f(b) < \alpha < f(a)$).

Then $\exists c \in (a,b)$ such that $f(c) = \alpha$.

Remarks:

1. Note the closed interval domain of f is a must to ensure $f(a)$ and $f(b)$ are defined.
2. If we set $\alpha \in \left[f(a), f(b)\right]$ (or $\alpha \in \left[f(b), f(a)\right]$), then $\exists c \in \left[a, b\right] \ni f(c) = \alpha$.
3. When $\alpha = 0$, $f(a)f(b) < 0$ and the implication is $\exists c \in \left(a, b\right) \ni f(c) = 0$. This is a corollary we will see later.

Proof. Without losing its generality, let $f(a) < f(b)$.

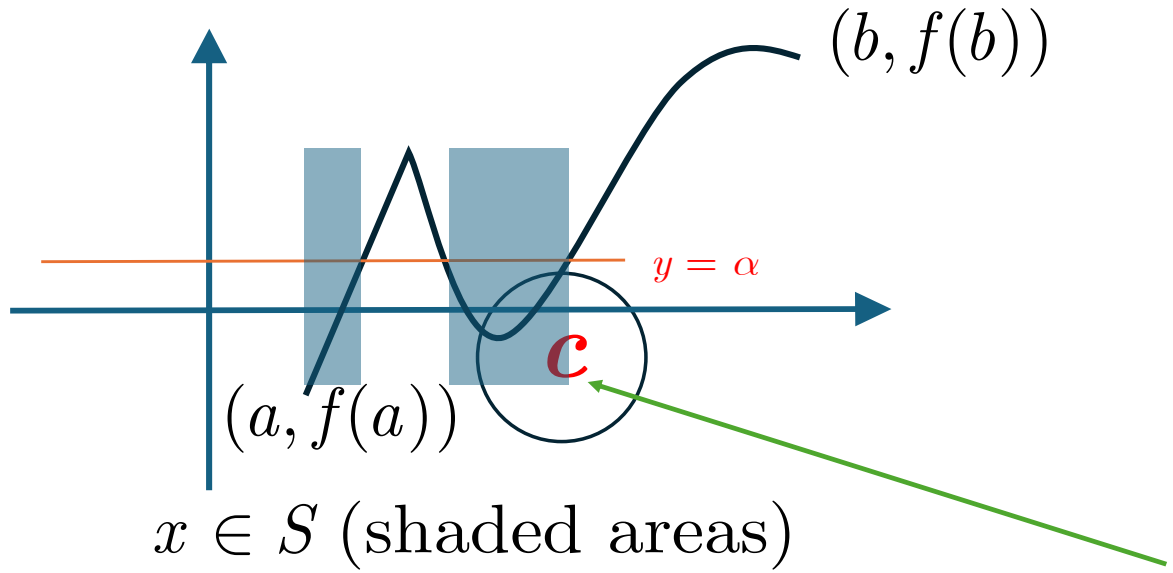
Let $\alpha \in \mathbb{R}$ such that $\alpha \in \Big(f(a), f(b)\Big)$.

Define the set $S = \Big\{x \in \Big[a, b\Big] : f(x) \leq \alpha\Big\}$

Then S is bounded above.

By the completeness axiom, $c = \sup S \in \mathbb{R}$ exists.

Furthermore, since $\left[a, b\right]$ is closed, $c \in S$, i.e., $c = \max S$.



Since $c = \sup S$, it follows by Theorem 4.2(i) that

$$\exists a \in S \ni c - \varepsilon < a \leq c \ \forall \varepsilon > 0 .$$

The statement above is true <u>for all $\varepsilon > 0$</u> :

When $\varepsilon = 1$, there exists an $a_1 \in S$ such that $c - 1 < a_1$.

When $\varepsilon = \frac{1}{2}$, there exists an $a_2 \in S$ such that $c - \frac{1}{2} < a_2$.

...

When $\varepsilon = \frac{1}{n}$, there exists an $a_n \in S$ such that $c - \frac{1}{n} < a_n$.

...

Meaning to say $\exists a_n \in S$ s.t. $c - \frac{1}{n} < a_n \leq c \; \forall n \in \mathbb{N}$.

Construct sequence $\left(a_n\right)$ taking all the above $a_n \in S$.

Note that a_n may not be distinct to a_m for any $n \neq m$ but that is fine because we are constructing a sequence.

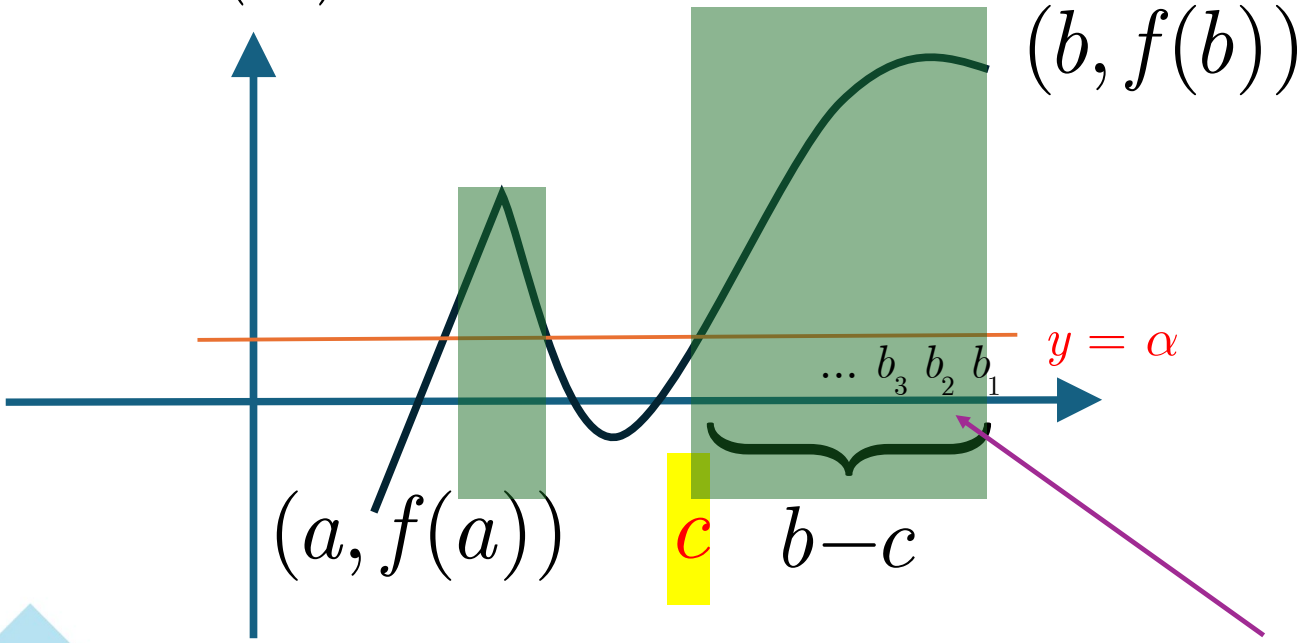
Note that also $c - \frac{1}{n} < a_n \leq c \; \forall n \in \mathbb{N}$.

By the squeeze theorem, $\lim\left(a_n\right) = c$.

Now we let $T = \left\{ x \in \left[a,b\right] : f(x) \geq \alpha \right\}$.

Define a sequence $b_n \in T$ by $b_n = c + \frac{b-c}{n}$.

Note that $\lim\left(b_n\right) = c$.



$x \in T$ (shaded areas)

Since $a_n \in S = \left\{x \in \left[a,b\right] : f(x) \leq \alpha\right\}$, $b_n \in T = \left\{x \in \left[a,b\right] : f(x) \geq \alpha\right\}$,

which means

$$\left(f(a_n)\right) = \left(f(a_1), f(a_2), f(a_3), \ldots\right) \quad \left(f(b_n)\right) = \left(f(b_1), f(b_2), f(b_3), \ldots\right)$$

$$\overset{\wedge|}{\alpha} \quad \overset{\wedge|}{\alpha} \quad \overset{\wedge|}{\alpha} \qquad\qquad \overset{\vee|}{\alpha} \quad \overset{\vee|}{\alpha} \quad \overset{\vee|}{\alpha}$$

By Theorem 6.3 (Chapter 2), $\boxed{\lim\left(f(a_n)\right) \leq \alpha}$ and $\boxed{\lim\left(f(b_n)\right) \geq \alpha}$.

f is continuous on $[a,b] \Leftrightarrow f$ is sequentially continuous on $[a,b]$.

$\lim(a_n) = \lim(b_n) = c \Rightarrow \lim(f(a_n)) = \lim(f(b_n)) = f(c)$.

From the last slide, $\lim(f(a_n)) \leq \alpha$ and $\lim(f(b_n)) \geq \alpha$.

Hence, $f(c) = \lim(f(a_n)) = \lim(f(b_n)) = \alpha$.

Next, we prove that $c \neq a, b$.

Prove by contradiction: Suppose $b = c$.

From previous results, we have

$\alpha \geq \lim\Big(f(a_n)\Big) = f(c) = f(b)$, a contradiction

since $\alpha \in \Big(f(a), f(b)\Big)$, i.e., $\alpha < f(b)$.

Similarly, we suppose $a = c$, then

$$f(a) = f(c) = \lim\Big(f(a_n)\Big) = \lim\Big(f(b_n)\Big) \geq \alpha\,,$$

contradict to the fact that

$\alpha \in \Big(f(a), f(b)\Big)$, i.e., $\alpha > f(a)$.

Thus, we conclude that when

$f : \left[a, b\right] \to \mathbb{R}$ is continuous on $\left[a, b\right]$,

and $\alpha \in \mathbb{R}$ is in between $f(a)$ and $f(b)$,

then $\exists c \in \left(a, b\right)$ such that $f(c) = \alpha$.

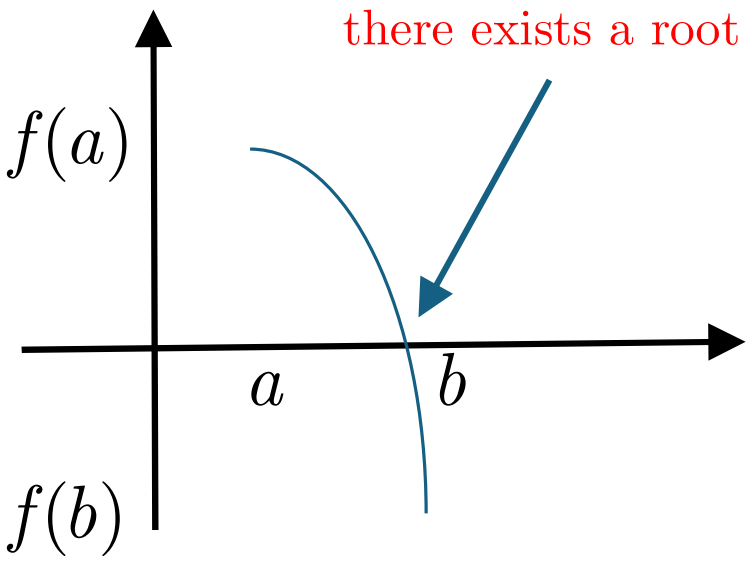
Corollary 8.4. Bolzano's theorem

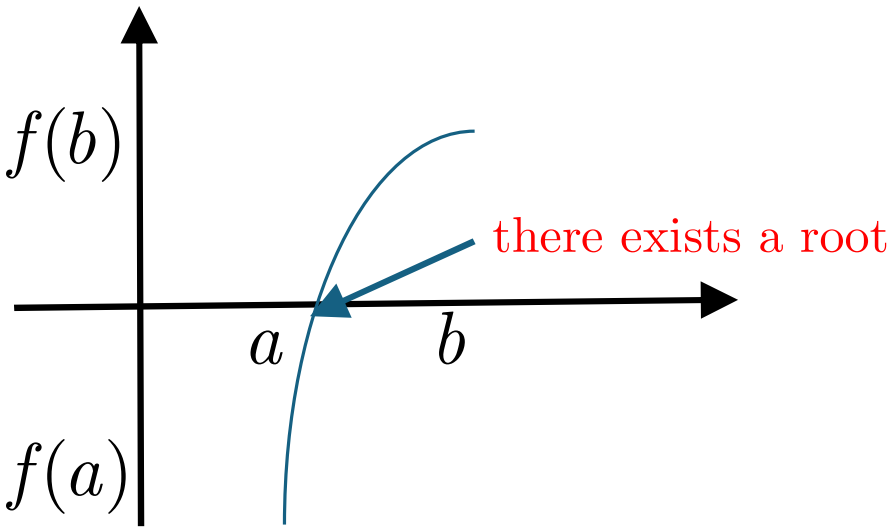
As an immediate result, we see that if $\alpha = 0$ (α is in between $f(a)$ and $f(b)$).

Then $f(a)$ and $f(b)$ must have different signs (one positive and one negative). So $f(a)f(b) < 0$.

By the IVT, $\exists c \in \left(a, b\right)$ such that $f(c) = \alpha = 0$,

i.e., f has (at least) a root in between $\left(a, b\right)$.


there exists a root
$f(a)$
$f(b)$
a
b


$f(b)$
there exists a root
a
b
$f(a)$